CREDIT CARD FRAUD DETECTION USING RANDOM FOREST

Machine Learning Classification Project | Student Submission

Dataset	credit_card_fraud_detection_2026.csv	**Task**	Binary classification
Records	100,000 initially inspected	**Target**	is_fraud (0 = genuine, 1 = fraud)

1. Problem Statement

The objective is to classify credit-card transactions as genuine or fraudulent using transaction, customer, location, and spending-behaviour features. The model should identify fraudulent activity while limiting incorrect alerts on genuine transactions.

2. Data Preprocessing

The dataset was inspected for data types, missing values, distributions, outliers, and relationships between features. Unnecessary identifier/time fields and post-outcome fields (such as fraud_type and fraud_amount_lost) were excluded from predictors to reduce leakage risk. Missing browser values were represented as “Unknown”; categorical columns were converted to numeric indicators using one-hot encoding. IQR-based capping was explored for selected numeric features. The row with a missing is_fraud value was removed. Features and target were separated, followed by an 80:20 train-test split.

3. Model Used

A RandomForestClassifier was trained on the encoded training data. Random Forest combines multiple decision trees to capture non-linear patterns and interactions. Hyperparameter tuning with GridSearchCV (5-fold cross-validation) was attempted.

4. Evaluation Results

Metric	Result
Accuracy	97.97%
Fraud precision (class 1)	0.00
Fraud recall (class 1)	0.00
Fraud F1-score (class 1)	0.00
Test support	20,000 transactions (19,594 genuine; 406 fraud)

Confusion matrix	Predicted genuine (0)	Predicted fraud (1)
Actual genuine (0)	19,594	0
Actual fraud (1)	406	0

Important interpretation: Although accuracy is 97.97%, the model predicted every test transaction as genuine and detected none of the 406 fraudulent transactions. Therefore, accuracy is misleading because the dataset is highly imbalanced; the current model is not effective for fraud detection.

5. Key Findings and Conclusion

Exploratory analysis showed class imbalance and some correlated numeric features. The current evaluation does not establish useful fraud-predictive features; feature importance should be examined after improving and re-evaluating the model. Recommended next steps are to use class_weight='balanced' or training-only resampling, tune the decision threshold, and compare models using fraud-class precision, recall, F1-score, and PR-AUC. The final model should be selected based on its ability to detect fraud, not accuracy alone. In conclusion, the Random Forest workflow was implemented, but the present model requires further improvement before it can be considered suitable for fraud detection.

Kaggle source: The exact Kaggle dataset URL was not provided; add the original Kaggle page link before submission. Confirm the final row count and ensure preprocessing steps are fitted using training data only.